

***CO 3 - ASSESSMENT TOOL 2: Data  
Interpretation***

***COURSE NAME: FUNDAMENTALS OF  
DATA SCIENCE***

***COURSE CODE: DSA0404***

***Name: YEDURURI ANJANI PRIYA***

***Reg No:192324176***

***Submitted to: Dr.Krishnaraj M***

### 1. **Data Summarization**

A dataset of 11 employee salaries (in ₹'000) is: 32, 35, 36, 38, 40, 42, 45, 48, 50, 95, 120.

- (a) Compute the mean and median.
- (b) Explain why the mean is a poor measure of central tendency here.
- (c) Which measure of dispersion would you trust more — standard deviation or IQR — and why?

### **SOLUTION:**

Dataset:

**32, 35, 36, 38, 40, 42, 45, 48, 50, 95, 120**

- (a) Mean and Median

**Mean:**

$$\begin{aligned}\bar{x} &= \frac{32 + 35 + 36 + 38 + 40 + 42 + 45 + 48 + 50 + 95 + 120}{11} \\ &= \frac{581}{11} = \boxed{52.82}\end{aligned}$$

So, **Mean = ₹52.82 thousand**

Since there are 11 observations, the middle value is the 6th value:

So, **Median = ₹42 thousand**

### **(b) Why is mean a poor measure here?**

The values **95 and 120** are unusually high compared with the other salaries. These extreme values pull the mean upward from the typical salary level.

Mean = ₹52.82 thousand

Median = ₹42 thousand

Therefore, the *median better represents the typical employee salary.*

**(c) Standard deviation or IQR?**

IQR is more trustworthy because the dataset contains extreme values/outliers.

Standard deviation is strongly affected by extreme observations.

IQR measures the spread of the middle 50% and is resistant to outliers.

Answer: IQR.

### ***QUE:2.OutlierTreatment***

***Using the dataset above, apply the  $1.5 \times IQR$  rule to identify outliers.***

***Show Q1, Q3, IQR, and the lower/upper fences. Which values qualify as outliers? If you cap them at the fences (Winsorization) instead of removing them, how would the mean change qualitatively?***

Sorted dataset:

32, 35, 36, 38, 40, 42, 45, 48, 50, 95, 120

Using the median-of-halves method:

Q1

Lower half:

32, 35, 36, 38, 40

***Q1=36***

Q3

Upper half:

45, 48, 50, 95, 120

***Q3=50***

***IQR***

$$IQR = Q3 - Q1$$

$$IQR = 50 - 36 = \boxed{14}$$

***Lower Fence***

$$\begin{aligned}
 &Q1 - 1.5(IQR) \\
 &= 36 - 1.5(14) \\
 &= 36 - 21 = \boxed{15}
 \end{aligned}$$

***Upper Fence***

$$\begin{aligned}
 &Q3 + 1.5(IQR) \\
 &= 50 + 1.5(14) \\
 &= 50 + 21 = \boxed{71}
 \end{aligned}$$

***Outliers***

Any value below 15 or above 71 is an outlier.

$95 > 71 \rightarrow \text{Outlier}$

$120 > 71 \rightarrow \text{Outlier}$

Therefore:

***Outliers=95,120***

***Winsorization***

If we cap the outliers at the upper fence of 71:

Original:

***32, 35, 36, 38, 40, 42, 45, 48, 50, 95, 120***

After capping:

***32, 35, 36, 38, 40, 42, 45, 48, 50, 71, 71***

New mean:

$$\frac{508}{11} = \boxed{46.18}$$

The mean changes from 52.82 to approximately 46.18.

**Qualitatively:** Winsorization reduces the influence of the extreme salaries, so the mean moves downward and closer to the median.

## 2. Measuring Asymmetry

Two datasets have the same mean (50) and standard deviation (10), but Dataset A has Pearson's skewness coefficient = +1.2 and Dataset B has -0.8. Sketch (describe) the shape of each distribution and state which is more likely to have salary-like data with a long right tail.

Both datasets have:

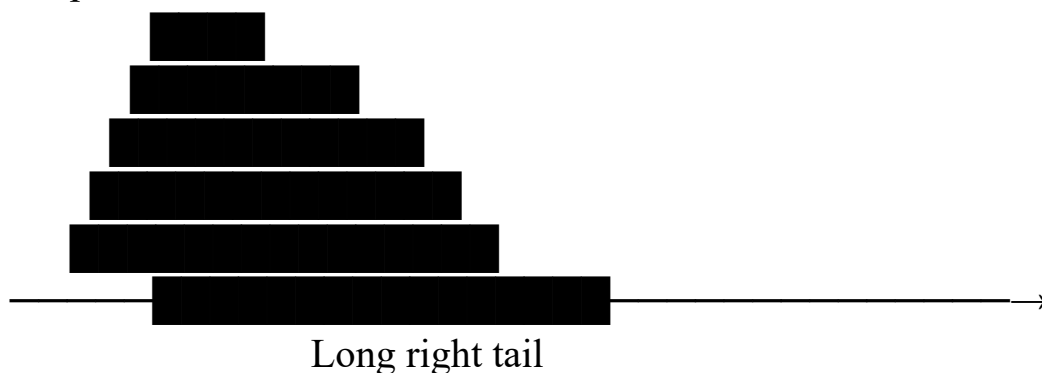
Mean = 50

Standard deviation = 10

Dataset A: Pearson's skewness = +1.2

A positive skewness means the distribution has a long tail toward the right.

Shape:



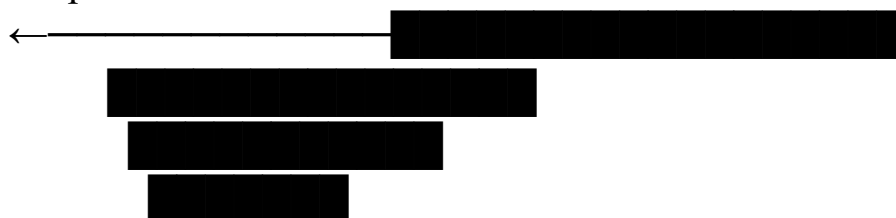
Most observations ← concentrated here long right tail

So Dataset A is right-skewed / positively skewed.

Dataset B: Pearson's skewness = -0.8

A negative skewness means the distribution has a long tail toward the left.

Shape:



Left tail

←—————| concentrated observations →

So Dataset B is left-skewed / negatively skewed.

Which is more like salary data?

Dataset A is more likely to represent salary data because salaries often have a long right tail: most employees earn moderate salaries while a smaller number of employees earn very high salaries.

### 3. *Coefficient of Skewness Calculation*

*For a dataset: Mean = 62, Median = 58, Mode = 52, SD = 12.*

*Calculate skewness using*

*(a) Karl Pearson's coefficient (mode method),*

*(b) Karl Pearson's coefficient (median method,  $Sk = 3(\text{Mean} - \text{Median})/SD$ ).*

*Compare the two results and comment on why they might differ slightly.*

Given:

Mean = 62

Median = 58

Mode = 52

SD = 12

*(a) Karl Pearson's coefficient — Mode method*

Formula:

$$Sk = \frac{\text{Mean} - \text{Mode}}{SD}$$

Substitute:

$$Sk = \frac{62 - 52}{12}$$

$$= \frac{10}{12}$$

$$Sk = 0.833$$

Therefore, the distribution is *positively skewed*.

**(b) Karl Pearson's coefficient — Median method**

Formula:

$$Sk = \frac{3(\text{Mean} - \text{Median})}{SD}$$

Substitute:

$$Sk = \frac{3(62 - 58)}{12}$$
$$= \frac{3(4)}{12}$$
$$= \frac{12}{12}$$
$$Sk = 1.0$$

**Comparison**

**Method** Mode method

|               | <b>Skewness</b> |
|---------------|-----------------|
|               | <b>0.833</b>    |
| Median method | <b>1.000</b>    |

Both indicate **positive/right skewness**.

They differ slightly because the two formulas use **different measures of central tendency**—the mode in the first formula and the median in the second. For a moderately skewed distribution, these estimates need not be identical.

**QUE: 5. Variance vs Standard Deviation Interpretation**

Two production lines report these stats:

Line A: Mean = 200 units/day, Variance = 64

Line B: Mean = 150 units/day, Variance = 36

Which line has greater relative variability? Use the Coefficient of Variation to justify your answer, not just raw variance.

|                 | <b>Line A</b>        | <b>Line B</b>        |
|-----------------|----------------------|----------------------|
| <b>Mean</b>     | <b>200 units/day</b> | <b>150 units/day</b> |
| <b>Variance</b> | <b>64</b>            | <b>36</b>            |

**First calculate standard deviation.**

**Line A**

**SD=64=8**

**Coefficient of Variation:**

$$CV = \frac{SD}{Mean} \times 100$$

$$CV = \frac{8}{200} \times 100$$

$$CV = 4\%$$

**Line B**

$$SD = \sqrt{36} = 6$$

$$CV = \frac{6}{150} \times 100$$

$$CV = 4\%$$

### **Conclusion**

Both production lines have the same relative variability:

$$CV_A = CV_B = 4\%$$

Although Line A has a higher raw variance (64 vs 36), it also has a higher mean (200 vs 150). Once variability is standardized relative to the mean using CV, both lines are equally variable.